

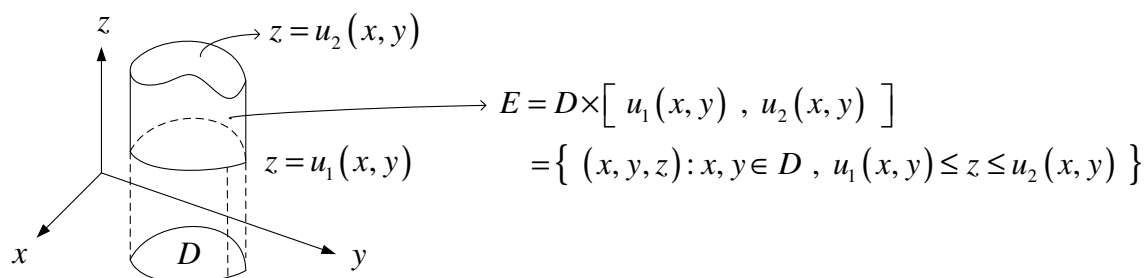
§15.7 Triple Integrals

Homework : 3,7,9,11,15,19,25,29,31,33

屋頂： $u = f(x, y, z)$

地基：在三維空間的物體E.

例如



Triple Integral :

$$\iiint_E f(x, y, z) dV = \iint_D \left[\int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz \right] dA \quad (1)$$

註：屋頂夠好，上述等式成立，事實上，

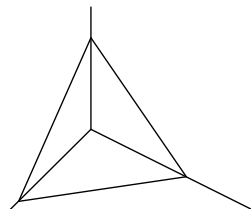
Fubini Theorem：屋頂夠好（沒三維破洞，或只有有限的二維或一維的裂縫），則 Triple Integral 可寫成遞迴式(iterated)的三個一維積分。

Example 1 : $\iiint_E z dV$, where E is the solid bounded by the four planes

$$x = 0, y = 0, z = 0 \text{ and } x + y + z = 1$$

Solution :

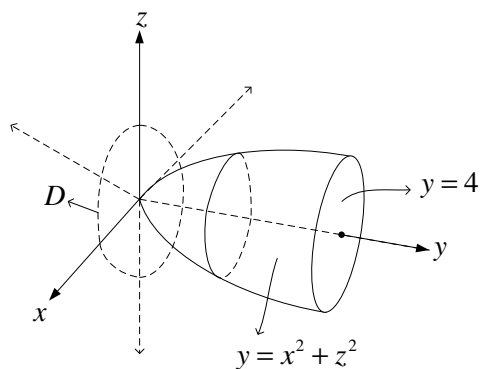
$$\begin{aligned} & \iiint_E z dV \\ &= \int_0^1 \int_0^{1-x} \int_0^{1-x-y} z dz dy dx \\ &= \frac{1}{24} \\ &= \int_0^1 \int_0^{1-z} \int_0^{1-y-z} z dx dy dz \end{aligned}$$



Example 2 : $\iiint_E \sqrt{x^2 + z^2} \, dV$. E is bounded by $\begin{cases} y = x^2 + z^2 \\ y = 4 \end{cases}$.

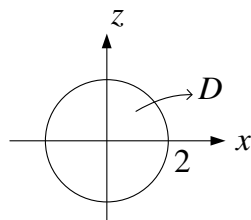
Solution :

地基 E



$$E = D \times [x^2 + z^2, 4]$$

$$D = x^2 + z^2 \leq 4$$



$$\Rightarrow \iiint_E \sqrt{x^2 + z^2} \, dV$$

$$= \iint_D \left(\int_{x^2+z^2}^4 \sqrt{x^2 + z^2} \, dy \right) dA$$

$$= \iint_{x^2+z^2 \leq 4} \left(\sqrt{x^2 + z^2} \right) (4 - x^2 - z^2) \, dA = \int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} (4 - x^2 - z^2) \sqrt{x^2 + z^2} \, dz dx$$

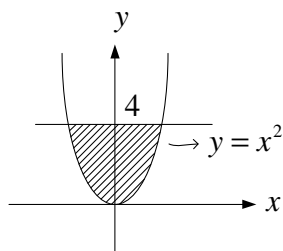
$$= \int_0^{2\pi} \int_0^2 r^2 (4 - r^2) \, dr d\theta$$

$$= \frac{128\pi}{15}$$

E 的另一看法 :

$$E = D' \times \left[-\sqrt{y-x^2}, \sqrt{y-x^2} \right]$$

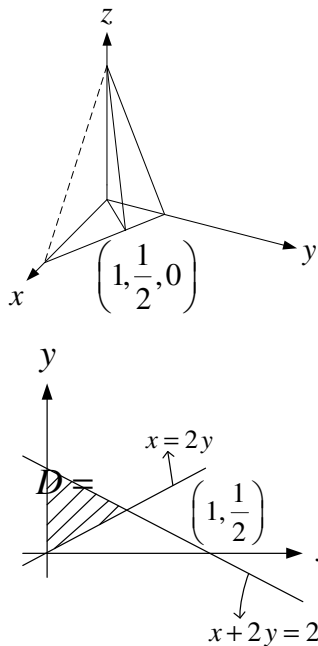
$D' =$



$$\begin{aligned}\text{Triple Integral} &= \iint_{D'} \int_{-\sqrt{y-x^2}}^{\sqrt{y-x^2}} \sqrt{x^2+z^2} \, dz dA \\ &= \int_{-2}^2 \int_{x^2}^4 \int_{-\sqrt{y-x^2}}^{\sqrt{y-x^2}} \sqrt{x^2+z^2} \, dz dy dx \text{ (不好算)}\end{aligned}$$

Example 3 : Use a triple integral to find the volume of the tetrahedron T bounded by the planes $x+2y+z=2$, $x=2y$, $x=0$ and $z=0$.

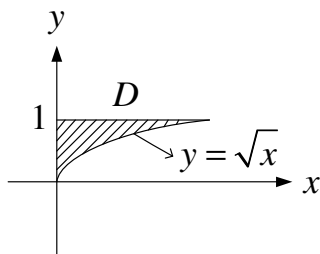
Solution :



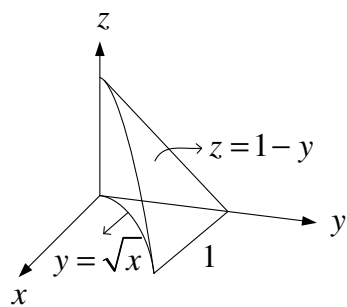
$$\begin{aligned}V &= \iiint_D 1 \, dz \, dA \\ &= \int_0^1 \int_{\frac{x}{2}}^{\frac{2-x}{2}} \int_0^{2-x-2y} 1 \, dz \, dy \, dx \\ &= \frac{1}{3}\end{aligned}$$

Example 4 : Rewrite $\int_0^1 \int_{\sqrt{x}}^1 \int_0^{1-y} f(x, y, z) \, dz dy dx$ as an equivalent iterated integral in the five other orders.

Solution :



$$\begin{aligned}E &= D \times [0, 1-y] \\ &= \{ (x, y, z) : (x, y) \in D \text{ and } 0 \leq z \leq 1-y \}\end{aligned}$$



$$\text{Triple Integral} = \iiint_D \int_0^{1-y} f \, dz \, dA$$

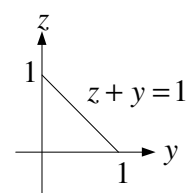
$$= \int_0^1 \int_0^{y^2} \int_0^{1-y} f \, dz \, dx \, dy$$

$$= \int_0^1 \int_0^{1-z} \int_0^{y^2} f \, dx \, dy \, dz \quad \longrightarrow$$

$$= \int_0^1 \int_0^{1-y} \int_0^{y^2} f \, dx \, dz \, dy \quad \nearrow$$

$$= \int_0^1 \int_0^{1-\sqrt{x}} \int_{\sqrt{x}}^{1-z} f \, dy \, dz \, dx$$

$$= \int_0^1 \int_0^{(1-z)^2} \int_{\sqrt{x}}^{1-z} f \, dy \, dx \, dz$$



$$\begin{cases} y = \sqrt{x} \\ z = 1 - y \end{cases} \Rightarrow z = 1 - \sqrt{x}$$

